

Image Classification Using Convolutional Neural Networks (CNNs) on the CIFAR-10 Dataset

Week 6 Internship Task – Computer Vision with CIFAR-10

AnalystLab Africa Generative AI Internship Program (Batch B)

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Problem Statement

The objective of this project was to develop an image classification model capable of accurately identifying and classifying images from the CIFAR-10 dataset into ten distinct object categories using Computer Vision techniques and Convolutional Neural Networks (CNNs).

Approach Used

The CIFAR-10 dataset was loaded and explored to understand its structure and image classes. Image preprocessing was performed by normalizing pixel values and one-hot encoding the target labels. A Convolutional Neural Network (CNN) consisting of convolutional, pooling, flatten, and dense layers was then developed, trained, and evaluated using the test dataset.

Model Performance

The CNN achieved a training accuracy of 76.51% and a test accuracy of 70.12%, with a test loss of 0.8874. The results indicate that the model successfully learned meaningful visual features and was able to generalize reasonably well to unseen images.

Key Observations and Lessons Learned

- CNNs are highly effective for image classification tasks due to their ability to automatically extract visual features from images.
- Image preprocessing, particularly normalization, plays a crucial role in improving model performance.

- The model demonstrated good generalization ability, with only mild signs of overfitting.
- Some image categories with similar visual characteristics were more challenging to classify accurately.
- Performance evaluation metrics such as accuracy curves, classification reports, and confusion matrices provide valuable insights into model behavior and prediction quality.